

Importing the Student Performance Dataset

Table.TransformColumnTypes(#"Promoted Headers",{"gender", type text}, {"race/ethnicity", type text}, {"parental level of education", type text}, {"lunch", type text}, {"test preparation course", type text}, {"math score", type number}, {"reading score", type number}, {"writing score", type number})

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
1	female	group B	bachelor's degree	standard	none	72	74	74
2	female	group C	some college	standard	completed	69	90	88
3	female	group B	master's degree	standard	none	90	95	93
4	male	group A	associate's degree	free/reduced	none	47	57	44
5	male	group C	some college	standard	none	76	78	75
6	female	group B	associate's degree	standard	none	71	83	78
7	female	group B	some college	standard	completed	88	95	92
8	male	group B	some college	free/reduced	none	40	43	39
9	male	group D	high school	free/reduced	completed	64	64	67
10	female	group B	high school	free/reduced	none	38	60	50
11	male	group C	associate's degree	standard	none	58	54	52
12	male	group D	associate's degree	standard	none	40	52	43
13	female	group B	high school	standard	none	65	81	73
14	male	group A	some college	standard	completed	78	72	70
15	female	group A	master's degree	standard	none	50	53	58
16	female	group C	some high school	standard	none	69	75	76
17	male	group C	high school	standard	none	88	89	86
18	female	group B	some high school	free/reduced	none	18	32	28
19	male	group C	master's degree	free/reduced	completed	46	42	46
20	female	group C	associate's degree	free/reduced	none	54	58	61
21	male	group D	high school	standard	none	66	69	63
22	female	group B	some college	free/reduced	completed	65	75	70
23	male	group D	some high school	standard	none	44	54	53
24	female	group C	some high school	standard	none	69	73	73
25	male	group B	bachelor's degree	free/reduced	completed	72	74	74

The Student Performance dataset was imported into Power BI and opened in Power Query. The dataset contains student information along with their math, reading and writing scores.

Cleaning Text Data

Table.TransformColumns(#"Trimmed Text",{"gender", Text.Clean, type text}, {"race/ethnicity", Text.Clean, type text}, {"parental level of education", Text.Clean, type text})

	parental level of education	lunch	test preparation course	math score	reading score	writing score
1	bachelor's degree	standard	none	72	74	74
2	some college	standard	completed	69	90	88
3	master's degree	standard	none	90	95	93
4	associate's degree	free/reduced	none	47	57	44
5	some college	standard	none	76	78	75
6	associate's degree	standard	none	71	83	78
7	some college	standard	completed	88	95	92
8	some college	free/reduced	none	40	43	39
9	high school	free/reduced	completed	64	64	67
10	high school	free/reduced	none	38	60	50
11	associate's degree	standard	none	58	54	52
12	associate's degree	standard	none	40	52	43
13	high school	standard	none	65	81	73
14	some college	standard	completed	78	72	70
15	master's degree	standard	none	50	53	58
16	some high school	standard	none	69	75	76
17	high school	standard	none	88	89	86
18	some high school	free/reduced	none	18	32	28
19	master's degree	free/reduced	completed	46	42	46
20	associate's degree	free/reduced	none	54	58	61
21	high school	standard	none	66	69	63
22	some college	free/reduced	completed	65	75	70
23	some college	standard	none	44	54	53
24	some high school	standard	none	69	73	73
25	bachelor's degree	free/reduced	completed	72	74	74

The text columns were cleaned using the Clean transformation. This was applied to remove unwanted characters and ensure that the categorical values were consistent.

Adding Student ID

Table.TransformColumns(#"Trimmed Text",{"gender", Text.Clean, type text}, {"race/ethnicity", Text.Clean, type text}, {"parental level of education", Text.Clean, type text}, {"lunch", Text.Clean, type text}, {"test preparation course", Text.Clean, type text}, {"math score", Text.Clean, type text}, {"reading score", Text.Clean, type text}, {"writing score", Text.Clean, type text})

gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
female	group B	bachelor's degree	standard	none	72	72	72
female	group C	some college	standard	completed	69	90	90
female	group B	master's degree	standard	none	90	95	95
male	group A	associate's degree	free/reduced	none	47	57	57
male	group C	some college	standard	none	76	78	78
female	group B	associate's degree	standard	none	71	83	83
female	group B	some college	standard	completed	88	95	95
male	group B	some college	free/reduced	none	40	43	43
male	group D	high school	free/reduced	completed	64	64	64
female	group B	high school	free/reduced	none	38	60	60
male	group C	associate's degree	standard	none	58	54	54
male	group D	associate's degree	standard	none	40	52	52
female	group B	high school	standard	none	65	81	81
male	group A	some college	standard	completed	78	72	72
female	group A	master's degree	standard	none	50	53	53
female	group C	some high school	standard	none	69	75	75
male	group C	high school	standard	none	88	89	89
female	group B	some high school	free/reduced	none	18	32	32
male	group C	master's degree	free/reduced	completed	46	42	42
female	group C	associate's degree	free/reduced	none	54	58	58
male	group D	high school	standard	none	66	69	69
female	group B	some college	free/reduced	completed	65	75	75
male	group D	some college	standard	none	44	54	54
female	group C	some high school	standard	none	69	73	73

An Index Column was added to the dataset to create a unique identifier for each student. The index starts from 1 and was later used as the Student ID.

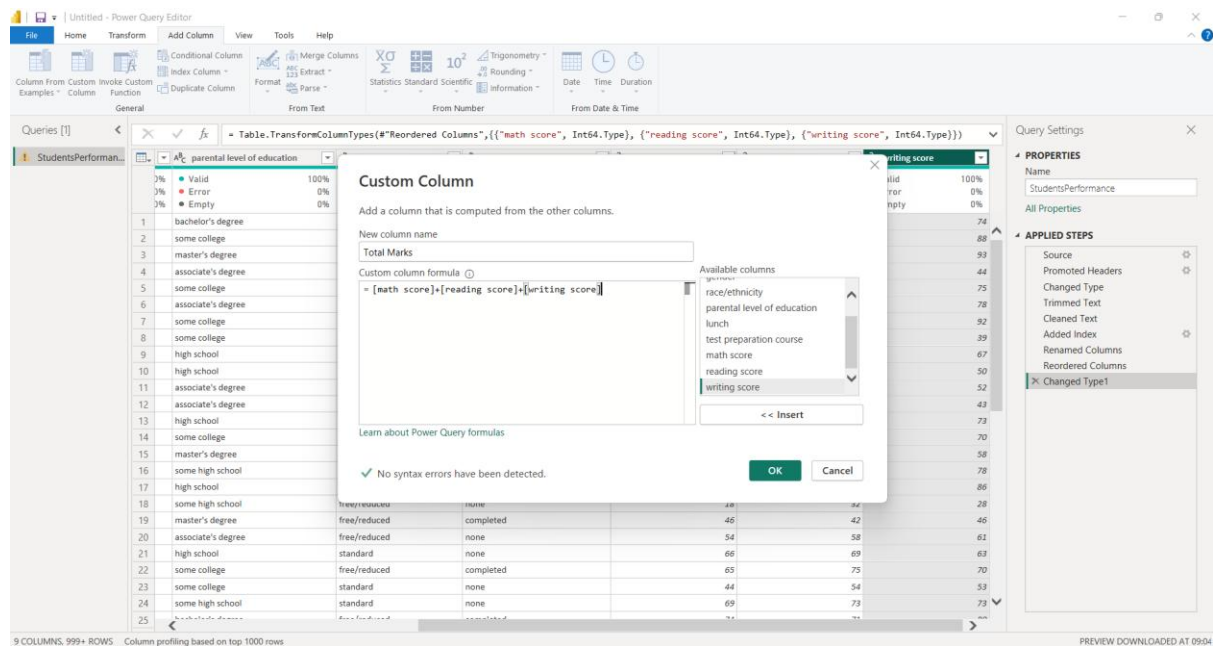
Renaming and Reordering Columns

Table.ReorderColumns(#"Renamed Columns",{"Student ID", "gender", "race/ethnicity", "parental level of education", "lunch", "test preparation course", "math score", "reading score", "writing score"})

Student ID	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
1	female	group B	bachelor's degree	standard	none	72	72	72
2	female	group C	some college	standard	completed	69	90	90
3	female	group B	master's degree	standard	none	90	95	95
4	male	group A	associate's degree	free/reduced	none	47	57	57
5	male	group C	some college	standard	none	76	78	78
6	female	group B	associate's degree	standard	none	71	83	83
7	female	group B	some college	standard	completed	88	95	95
8	male	group B	some college	free/reduced	none	40	43	43
9	male	group D	high school	free/reduced	completed	64	64	64
10	female	group B	high school	free/reduced	none	38	60	60
11	male	group C	associate's degree	standard	none	58	54	54
12	male	group D	associate's degree	standard	none	40	52	52
13	female	group B	high school	standard	none	65	81	81
14	male	group A	some college	standard	completed	78	72	72
15	female	group A	master's degree	standard	none	50	53	53
16	female	group C	some high school	standard	none	69	75	75
17	male	group C	high school	standard	none	88	89	89
18	female	group B	some high school	free/reduced	none	18	32	32
19	male	group C	master's degree	free/reduced	completed	46	42	42
20	female	group C	associate's degree	free/reduced	none	54	58	58
21	male	group D	high school	standard	none	66	69	69
22	female	group B	some college	free/reduced	completed	65	75	75
23	male	group D	some college	standard	none	44	54	54
24	female	group C	some high school	standard	none	69	73	73

The index column was renamed as Student ID and the columns were reordered to keep the student information and score columns in a proper sequence.

Creating Total Marks



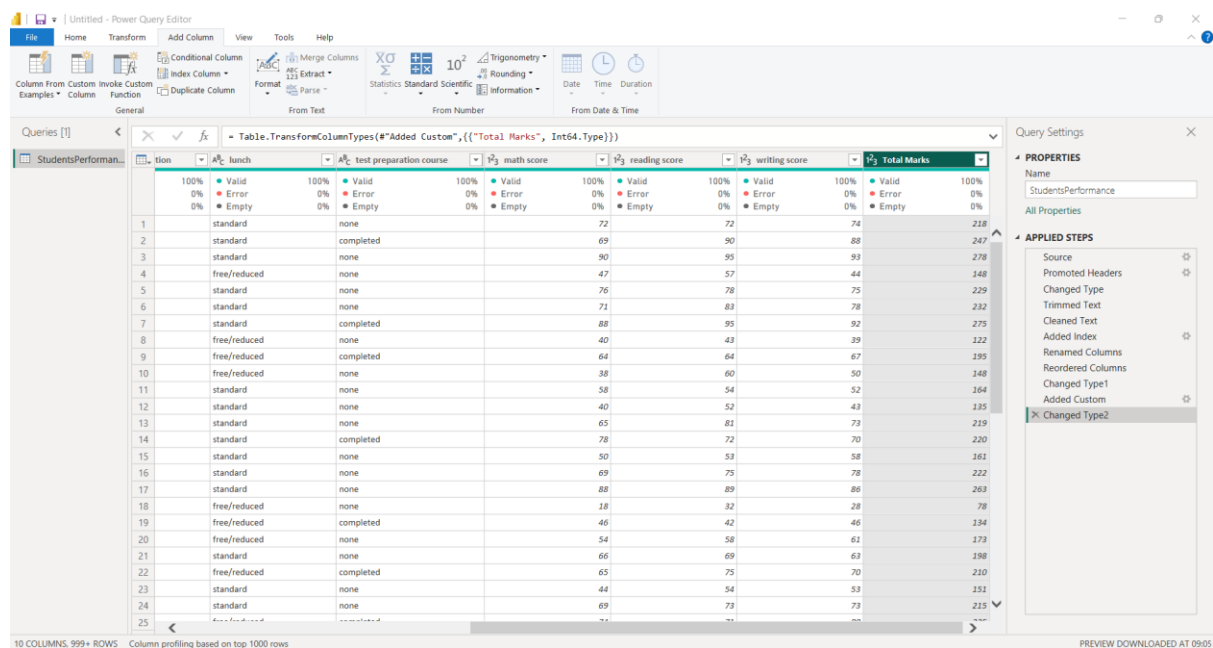
A custom column named Total Marks was created by adding the three subject scores.

The formula used was:

`[Math score] + [reading score] + [writing score]`

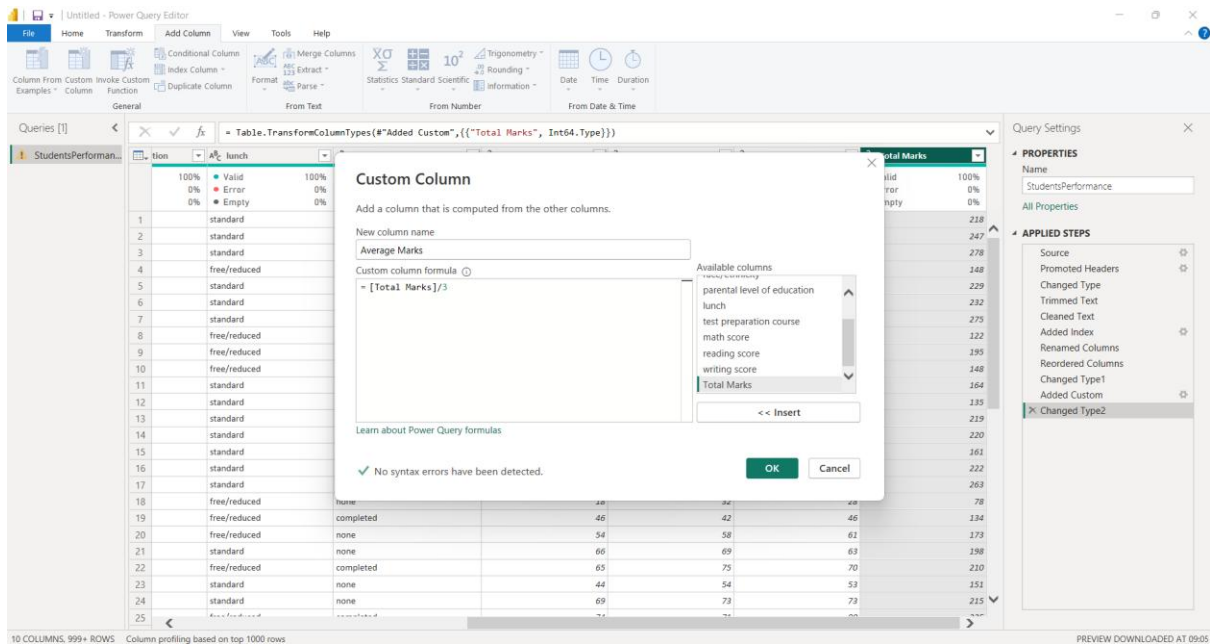
This gives the total marks obtained by each student across the three subjects.

Total Marks Column



The Total Marks column was added successfully to the dataset. It contains the combined score of math, reading and writing for each student.

Creating Average Marks



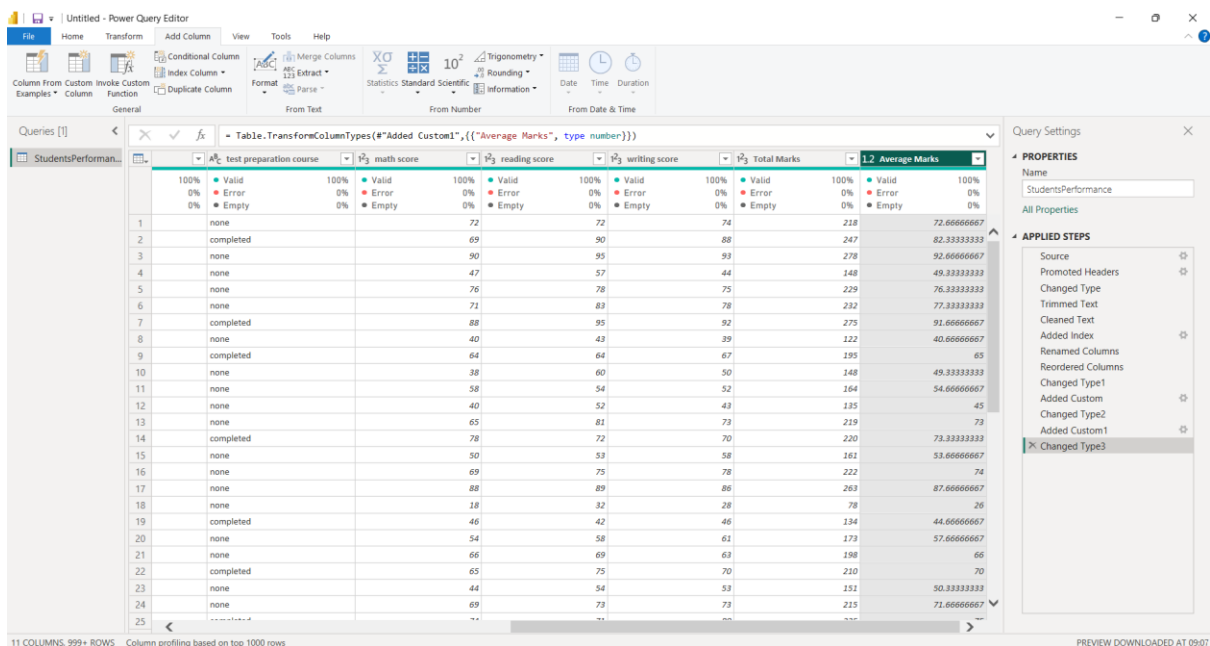
A custom column named Average Marks was created by dividing the Total Marks by 3.

The formula used was:

$[Total\ Marks] / 3$

This calculates the average score of each student across the three subjects.

Average Marks Column



The Average Marks column was added to the dataset. The values represent the average performance of each student across math, reading and writing.

Creating Grade

The screenshot shows the Power Query Editor interface. The 'Add Conditional Column' dialog box is open, allowing the user to create a new column named 'Grade' based on the 'Average Marks' column. The dialog includes a table of conditions to define the grade based on the average marks.

Condition	Operator	Value	Output
If Average Marks	is greater than or...	80	A
Else If Average Marks	is greater than or...	60	B
Else If Average Marks	equals	40	C
Else			F

The background data table shows the following columns: test preparation course, math score, reading score, writing score, Total Marks, and Average Marks. The data is as follows:

test preparation course	math score	reading score	writing score	Total Marks	Average Marks
none	66	69	63	198	66
completed	65	75	70	210	70
none	44	54	53	151	50.33333333
none	69	73	73	215	71.66666667

A conditional column named **Grade** was created based on Average Marks.

The grading conditions used were:

- Average Marks $\geq$ 80 $\rightarrow$ **A**
- Average Marks $\geq$ 60 $\rightarrow$ **B**
- Average Marks = 40 $\rightarrow$ **C**
- Otherwise $\rightarrow$ **F**

This column categorizes students based on their average performance.

Grade Column

Query: StudentsPerformance

Table.AddColumn(#"Changed Type3", "Grade", each if [Average Marks] >= 80 then "A" else if [Average Marks] >= 60 then "B" else if [Average Marks] >= 40 then "C" else "D")

Row	math score	reading score	writing score	Total Marks	Average Marks	Grade
1	72	72	74	218	72.66666667	B
2	69	90	88	247	82.33333333	A
3	90	95	93	278	92.66666667	A
4	47	57	44	148	49.33333333	F
5	76	78	75	229	76.33333333	B
6	71	83	78	232	77.33333333	B
7	88	95	92	275	91.66666667	A
8	40	43	39	122	40.66666667	F
9	64	64	67	195	65.33333333	B
10	38	60	50	148	49.33333333	F
11	58	54	52	164	54.66666667	F
12	40	52	43	135	45.33333333	F
13	65	81	73	219	73.0	B
14	78	72	70	220	73.33333333	B
15	50	53	58	161	53.66666667	F
16	69	75	78	222	74.0	B
17	88	89	86	263	87.66666667	A
18	18	32	28	78	26.66666667	F
19	46	42	46	134	44.66666667	F
20	54	58	61	173	57.66666667	F
21	66	69	63	198	66.0	B
22	65	75	70	210	70.0	B
23	44	54	53	151	50.33333333	F
24	69	73	73	215	71.66666667	B

The Grade column was added successfully. Each student is assigned a grade based on their calculated Average Marks.

Creating Pass/Fail Result

Table.AddColumn(#"Changed Type3", "Result", each if [Average Marks] >= 40 then "Pass" else "Fail")

Row	math score	reading score	writing score	Total Marks	Average Marks	Grade	Result
1	72	72	74	218	72.66666667	B	Pass
2	69	90	88	247	82.33333333	A	Pass
3	90	95	93	278	92.66666667	A	Pass
4	47	57	44	148	49.33333333	F	Fail
5	76	78	75	229	76.33333333	B	Pass
6	71	83	78	232	77.33333333	B	Pass
7	88	95	92	275	91.66666667	A	Pass
8	40	43	39	122	40.66666667	F	Fail
9	64	64	67	195	65.33333333	B	Pass
10	38	60	50	148	49.33333333	F	Fail
11	58	54	52	164	54.66666667	F	Fail
12	40	52	43	135	45.33333333	F	Fail
13	65	81	73	219	73.0	B	Pass
14	78	72	70	220	73.33333333	B	Pass
15	50	53	58	161	53.66666667	F	Fail
16	69	75	78	222	74.0	B	Pass
17	88	89	86	263	87.66666667	A	Pass
18	18	32	28	78	26.66666667	F	Fail
19	46	42	46	134	44.66666667	F	Fail
20	54	58	61	173	57.66666667	F	Fail
21	66	69	63	198	66.0	B	Pass
22	65	75	70	210	70.0	B	Pass
23	44	54	53	151	50.33333333	F	Fail
24	69	73	73	215	71.66666667	B	Pass

A conditional column named **Result** was created using Average Marks.

The condition used was:

- Average Marks $\geq 40 \rightarrow$ **Pass**
- Otherwise $\rightarrow$ **Fail**

This was used to identify whether each student met the minimum passing requirement.

Result Column

Query Settings

NAME: StudentsPerformance

APPLIED STEPS:

- Source
- Promoted Headers
- Changed Type
- Trimmed Text
- Cleaned Text
- Added Index
- Renamed Columns
- Reordered Columns
- Changed Type1
- Added Custom
- Changed Type2
- Added Custom1
- Changed Type3
- Added Conditional Column1
- Added Conditional Column1

	1.2 reading score	1.2 writing score	1.2 Total Marks	1.2 Average Marks	1.2 Grade	1.2 Result
1	72	72	74	218	72.66666667 B	Pass
2	69	90	88	247	82.33333333 A	Pass
3	90	95	93	278	92.66666667 A	Pass
4	47	57	44	148	49.33333333 F	Pass
5	76	78	75	229	76.33333333 B	Pass
6	71	83	78	232	77.33333333 B	Pass
7	88	95	92	275	91.66666667 A	Pass
8	40	43	39	122	40.66666667 F	Pass
9	64	64	67	195	65 B	Pass
10	38	60	50	148	49.33333333 F	Pass
11	58	54	52	164	54.66666667 F	Pass
12	40	52	43	135	45 F	Pass
13	65	81	73	219	73 B	Pass
14	78	72	70	220	73.33333333 B	Pass
15	50	53	58	161	53.66666667 F	Pass
16	69	75	78	222	74 B	Pass
17	88	89	86	263	87.66666667 A	Pass
18	18	32	28	78	26 F	Fail
19	46	42	46	134	44.66666667 F	Pass
20	54	58	61	173	57.66666667 F	Pass
21	66	69	63	198	66 B	Pass
22	65	75	70	210	70 B	Pass
23	44	34	33	151	50.33333333 F	Pass
24	69	73	73	215	71.66666667 B	Pass

The Result column was added successfully. Students with an Average Marks value of 40 or above are marked as Pass, while the remaining students are marked as Fail.

Selecting Student Marks Columns

Query Settings

NAME: StudentMarks

APPLIED STEPS:

- Source
- Promoted Headers
- Changed Type
- Trimmed Text
- Cleaned Text
- Added Index
- Renamed Columns
- Reordered Columns
- Changed Type1
- Added Custom
- Changed Type2
- Added Custom1
- Changed Type3
- Added Conditional Column1
- Added Conditional Column1
- Removed Other Columns

	1.2 reading score	1.2 writing score	1.2 math score	1.2 Student ID
1	72	72	74	1
2	69	90	88	2
3	90	95	93	3
4	47	57	44	4
5	76	78	75	5
6	71	83	78	6
7	88	95	92	7
8	40	43	39	8
9	64	64	67	9
10	38	60	50	10
11	58	54	52	11
12	40	52	43	12
13	65	81	73	13
14	78	72	70	14
15	50	53	58	15
16	69	75	78	16
17	88	89	86	17
18	18	32	28	18
19	46	42	46	19
20	54	58	61	20
21	66	69	63	21
22	65	75	70	22
23	44	34	33	23
24	69	73	73	24
25	74	71	80	25

A separate query named StudentMarks was created by keeping only the columns required for score analysis:

- Student ID
- Math Score
- Reading Score
- Writing Score

This separates the subject score information from the other student details.

Unpivoting Score Columns

The screenshot shows the Power Query Editor interface. The main area displays a table with four columns: 'Student ID', 'math score', 'reading score', and 'writing score'. The 'math score', 'reading score', and 'writing score' columns are selected. A context menu is open over these columns, and the 'Unpivot Columns' option is highlighted. The 'APPLIED STEPS' pane on the right shows the sequence of transformations applied to the data, including 'Unpivot Columns'.

Student ID	math score	reading score	writing score
1	72	72	72
2	69	90	90
3	90	95	95
4	47	57	57
5	76	78	78
6	71	83	83
7	88	95	95
8	40	43	43
9	64	64	64
10	38	60	60
11	58	54	54
12	40	52	52
13	65	81	81
14	78	72	72
15	50	53	53
16	69	75	75
17	88	89	89
18	18	32	28
19	46	42	46
20	54	58	61
21	66	69	63
22	65	75	70
23	44	54	53
24	69	73	73
25	74	71	80

The three subject score columns were selected and the Unpivot Columns option was used. This converts the separate subject columns into rows, making the data easier to analyse by subject.

Renaming Unpivoted Columns

The screenshot displays the Power Query Editor interface. The main area shows a table with three columns: 'Student ID', 'Subject', and 'Score'. The 'Student ID' column contains values from 1 to 25. The 'Subject' column contains values like 'math score', 'reading score', and 'writing score'. The 'Score' column contains numerical values ranging from 40 to 95. The 'Applied Steps' pane on the right lists the transformations applied to the data, including 'Source', 'Promoted Headers', 'Changed Type', 'Trimmed Text', 'Cleaned Text', 'Added Index', 'Renamed Columns', 'Reordered Columns', 'Changed Type1', 'Added Custom', 'Changed Type2', 'Added Custom1', 'Changed Type3', 'Added Conditional Column', 'Added Conditional Column1', 'Removed Other Columns', 'Unpivoted Columns', and 'Renamed Columns1'.

Student ID	Subject	Score
1	math score	72
2	reading score	72
3	writing score	74
4	math score	69
5	reading score	90
6	writing score	88
7	math score	90
8	reading score	95
9	writing score	93
10	math score	47
11	reading score	57
12	writing score	44
13	math score	76
14	reading score	78
15	writing score	75
16	math score	71
17	reading score	83
18	writing score	78
19	math score	88
20	reading score	95
21	writing score	92
22	math score	40
23	reading score	43
24	writing score	39
25	math score	64

After unpivoting, the Attribute column was renamed to Subject and the Value column was renamed to Score.

The resulting StudentMarks table contains:

- Student ID
- Subject
- Score

This creates a more flexible structure for analysing scores across different subjects.

Creating StudentDetails

Table: RemoveColumns(#"Added Conditional Column1",{"math score", "reading score", "writing score", "Total Marks", "Average Marks", "Grade", "Score"})

Student ID	gender	race/ethnicity	parental level of education	lunch	test preparation course
1	female	group B	bachelor's degree	standard	none
2	female	group C	some college	standard	completed
3	female	group B	master's degree	standard	none
4	male	group A	associate's degree	free/reduced	none
5	male	group C	some college	standard	none
6	female	group B	associate's degree	standard	none
7	female	group B	some college	standard	completed
8	male	group B	some college	free/reduced	none
9	male	group D	high school	free/reduced	completed
10	female	group B	high school	free/reduced	none
11	male	group C	associate's degree	standard	none
12	male	group D	associate's degree	standard	none
13	female	group B	high school	standard	none
14	male	group A	some college	standard	completed
15	female	group A	master's degree	standard	none
16	female	group C	some high school	standard	none
17	male	group C	high school	standard	none
18	female	group B	some high school	free/reduced	none
19	male	group C	master's degree	free/reduced	completed
20	female	group C	associate's degree	free/reduced	none
21	male	group D	high school	standard	none
22	female	group B	some college	free/reduced	completed
23	male	group D	some college	standard	none
24	female	group C	some high school	standard	none
25	male	group D	bachelor's degree	free/reduced	completed

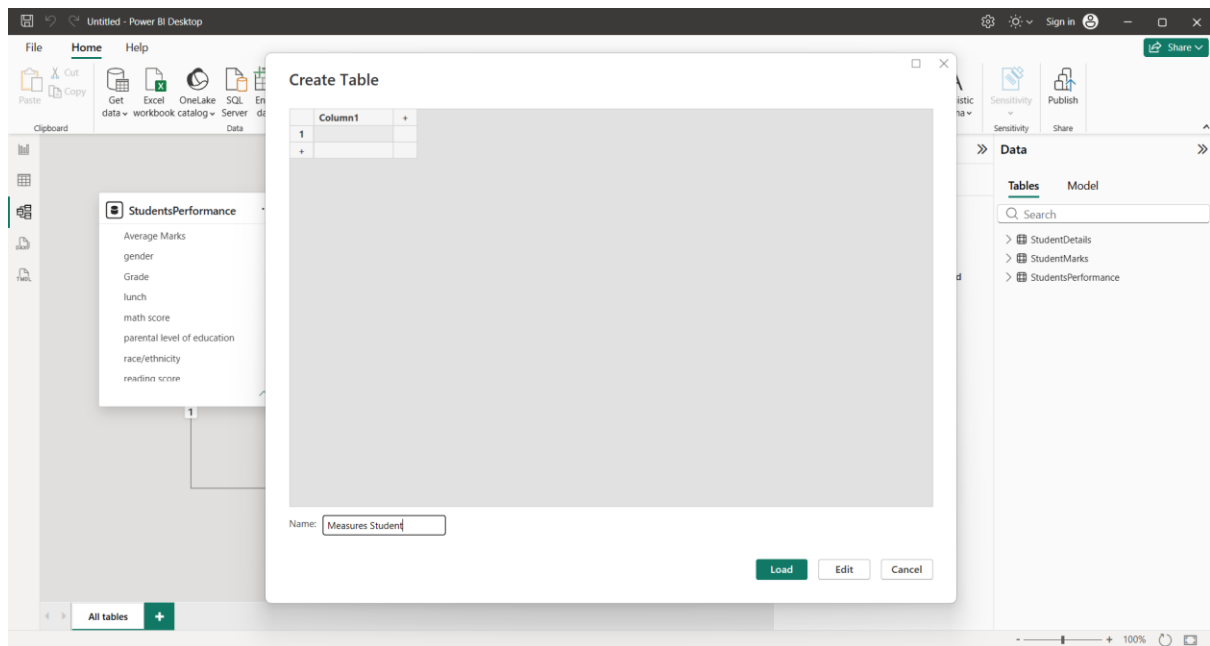
A separate StudentDetails table was created by removing the score-related and calculated columns from the original table.

The table retains the main student information such as:

- Student ID
- Gender
- Race/Ethnicity
- Parental Level of Education
- Lunch
- Test Preparation Course

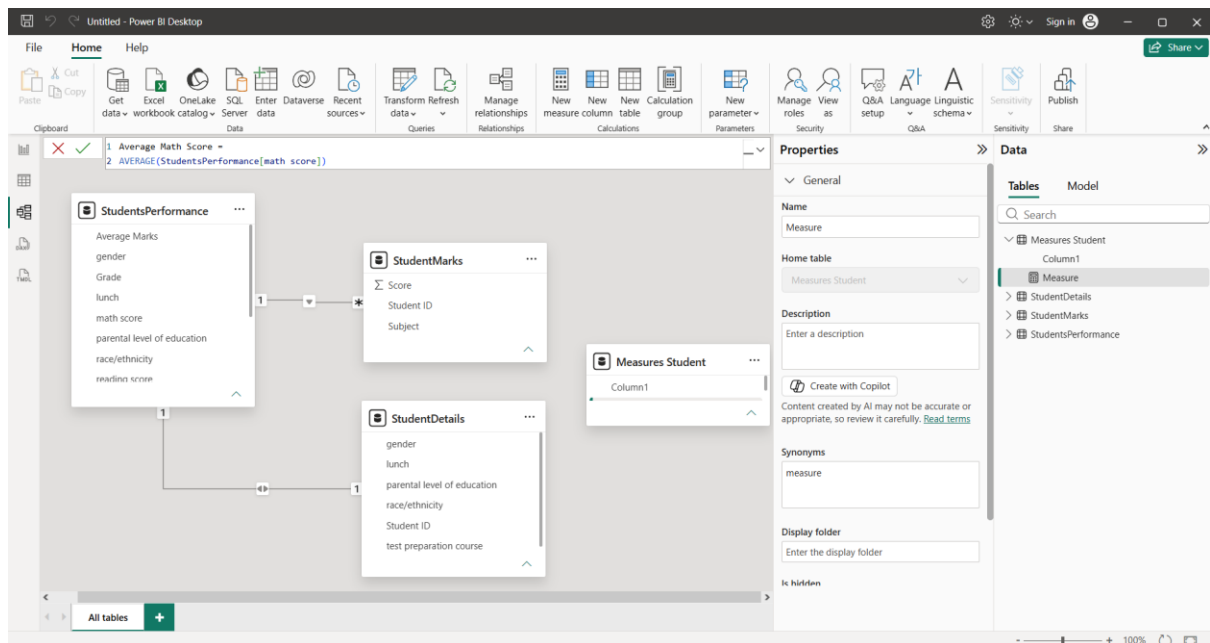
This separates student details from the marks data.

Creating Measures Student Table



A separate table named Measures Student was created to store the DAX measures. Keeping the measures in a separate table helps organize the calculations used for analysis.

Average Math Score



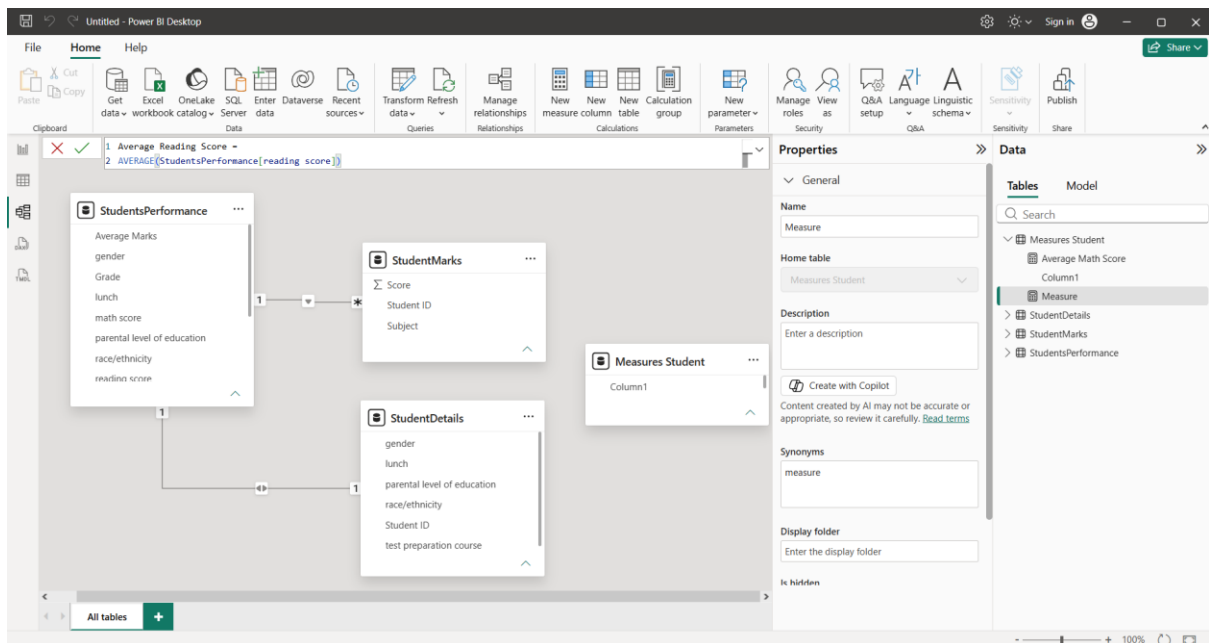
The Average Math Score measure was created using the AVERAGE function.

Average Math Score =

AVERAGE(StudentsPerformance[math score])

This measure calculates the average math score of the students.

Average Reading Score



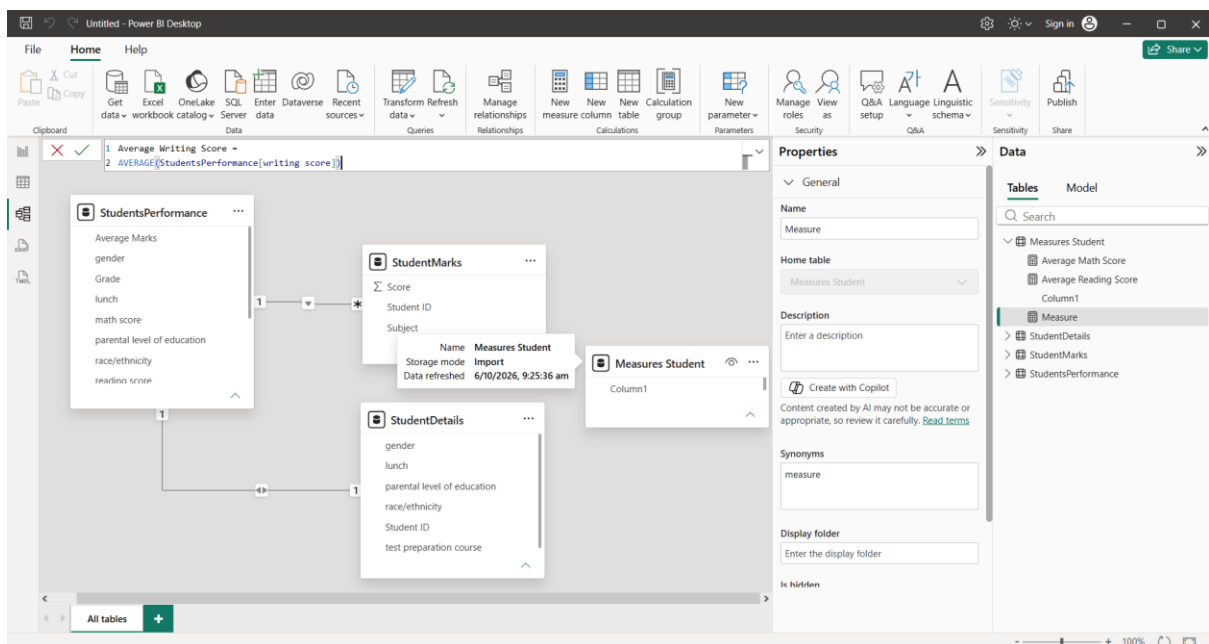
The Average Reading Score measure was created using the AVERAGE function.

Average Reading Score =

`AVERAGE(StudentsPerformance[reading score])`

This measure calculates the average reading score of the students.

Average Writing Score



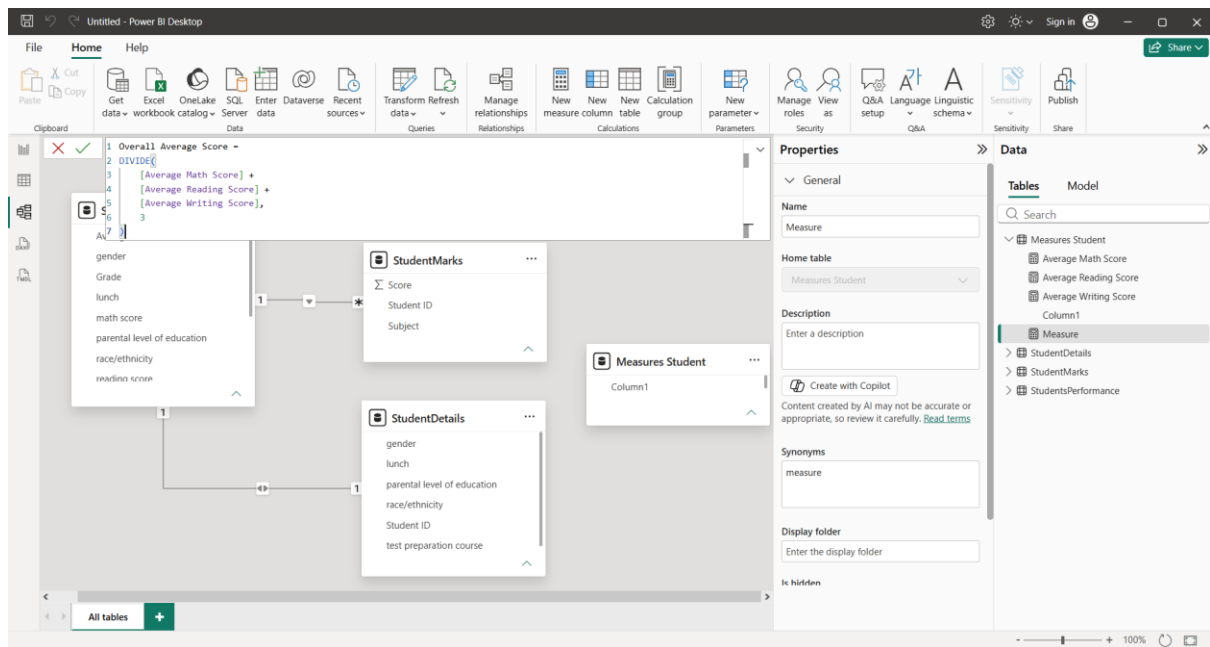
The Average Writing Score measure was created using the AVERAGE function.

Average Writing Score =

AVERAGE(StudentsPerformance[writing score])

This measure calculates the average writing score of the students.

Overall Average Score



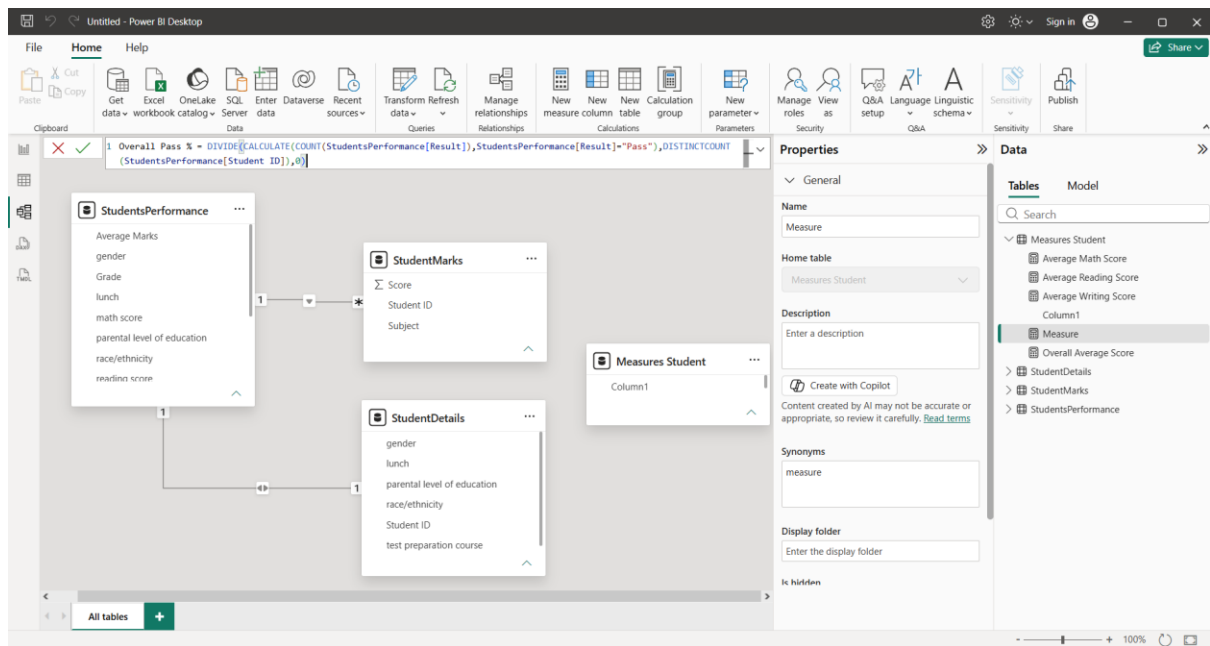
The Overall Average Score measure was created by taking the average of the three subject-level average scores.

Overall Average Score =

DIVIDE(
[Average Math Score] +
[Average Reading Score] +
[Average Writing Score],
3
)

This provides an overall average score across math, reading and writing.

Overall Pass %



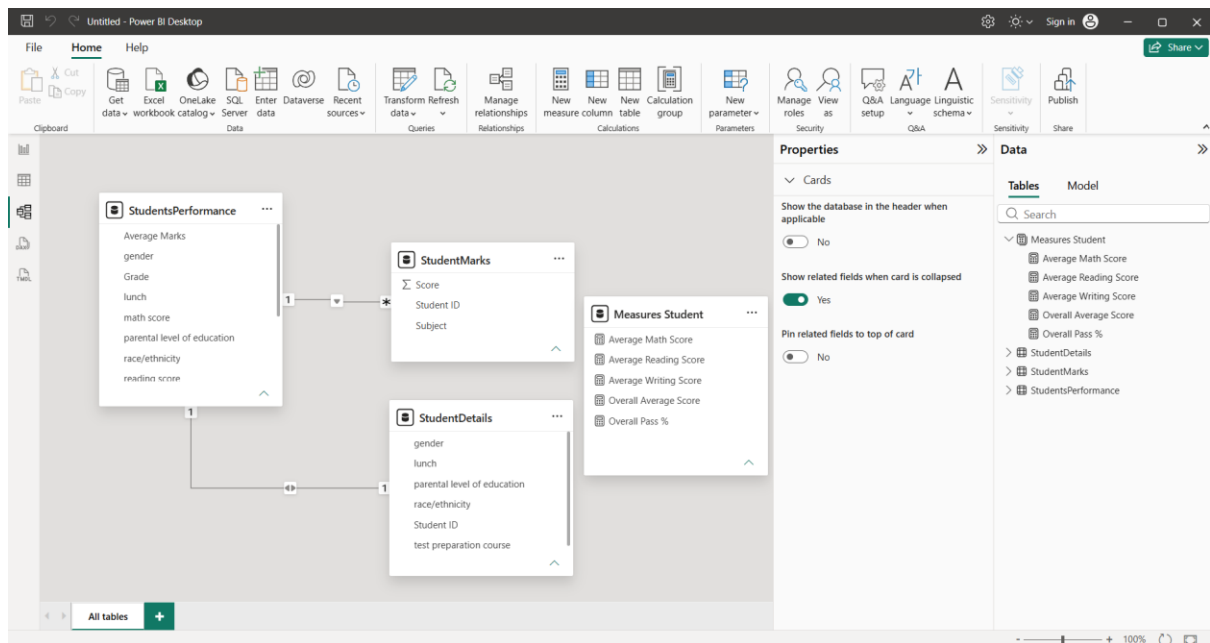
The Overall Pass % measure was created to calculate the percentage of students whose result is marked as Pass.

Overall Pass % =

```
DIVIDE(  
    CALCULATE(  
        COUNT(StudentsPerformance[Result]),  
        StudentsPerformance[Result] = "Pass"  
    ),  
    DISTINCTCOUNT(StudentsPerformance[Student ID]),  
    0  
)
```

This measure compares the number of students who passed with the total number of students.

Final Data Model



The final Power BI model contains the main StudentsPerformance table along with the StudentMarks, StudentDetails and Measures Student tables.

The StudentMarks table contains the subject-wise score information, while StudentDetails contains the main student information. The Measures Student table stores the DAX measures created for analysis.

The final model provides a structured way to analyse student performance and calculate overall academic metrics.